| UCE725: ADVANCED CONSTRUCTION MATERIALS AND TECHNIQUES | | | | |
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| | **3** | **1** | **0** | **3.5** |

**Course Objective**: The basic objective of the course is to expose the students to the latest and advanced construction materials used for thermal and sound insulation and special concretes used for specific field applications. The students will also be introduced to newer and latest construction techniques followed in construction industry

**Advanced Construction Materials:** Plastics, Timber products and Preservation, materials for thermal insulation, materials for sound insulation

**Special Concretes**: Light Weight Concrete, Vaccuum Concrete, Waste Material Based Concrete, Fiber reinforced concrete, Polymer Concrete Composites, Ferrocement, Concreting at High and Low Temperatures, Self- Compacting Concrete (SCC), Ready Mixed Concrete (RMC) and its characteristics and advantages, Shotcrete and concreting in tunnels

**Techniques for Tunneling and Formwork***:* Earthwork including cut and cover method, TBM, EBM and trenchless technology, Slip Form Shuttering, Latest type of Formwork, e.g. DOKA

**High Rise Structures:** Construction techniques for high rise buildings, chimneys. Special problems of high-rise construction

**Fire Resistance in Structures***:* Fire hazards in buildings and preventive measures

**Low Cost Housing:** Types, Design and advantages

**Course Learning Objectives (CLO)**

Upon the completion of this course, the students will be able to:

1. Characterize and specify advanced construction materials for thermal and sound insulation, smart materials and plastic and timber products.
2. Identify Special Concretes used in construction industry for specific applications.
3. Identify and Specify construction techniques for earthwork, tunneling and formwork.
4. Identify the various construction techniques for High Rise Buildings.

**Text Books:**

1. M.L. Gambhir , Neha Jamwal, Building Materials, Products, properties and systems, Mc Graw Hill (2011)
2. M.L. Gambhir, Concrete Technology, McGrawHill (2013)
3. Subir  Sarkar, SubhajitSarawati, Construction Technology, Oxford University Press (2008)

***Reference Books/Journals:***

*1.* Low Cost Houses, Publications by HUDCO, India Habitat Centre, Lodhi Road, New Delhi

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| (1982)<br>2. F. Glower, Structural Pre-cast Concrete, Oxford Publishers (1974)<br>3. Neil Jackson and R. K. Dhir, Civil Engineering materials, Macmillan Fourth edition (1996) |

**Evaluation Scheme:**

| Sr. No. | Evaluation Elements | Weights (%) |
|---|---|---|
| 1. | MST | 30 |
| 2. | EST | 50 |
| 3. | Sessionals (May include Assignments/Projects/Quizzes/Tutorials/Lab Evaluations) | 20 |